

Climate 5000 Series - Gen 4

24K BTU Single Zone Regular System

Wall Mounted IDU

Engineering
 Submittal
 Sheet



BOSCH

Overview and Project Information



Project name	
Location	
Architect	
Engineer	
General contractor	
Reference #	

24K Btu Single Zone Regular System - Wall Mounted IDU

Indoor part number		8-733-962-675
Indoor model number		BMS500-AAU024-1AHWXD
Outdoor part number		8-733-962-687
Outdoor model number		BMS500-AAS024-1CSXRD
Efficiency Data		AHRI 210/240 2023
SEER2		23
EER2		14.0
HSPF2 Region IV		11.7
HSPF2 Region V		8.7
COP(47F)		3.96
COP (5F)		2.54
Performance Data		AHRI 210/240 2023
Cooling	Capacity Range	5500-29500
	Rated Capacity	24000
	Current	7.4
	Input Power	1714
Heating	Capacity Range	9900-30800
	Rated Capacity (47F)	25000
	Rated Capacity (5F)	19600
	Current	8.3
	Input Power	1852
Electrical Data		
Power supply (V / Ø / Hz)		208-230 / 1 / 60
Min. circuit ampacity indoor/outdoor (A)		3.0 / 24.0
Max. fuse indoor/outdoor (A)		15.0 / 25.0
Compressor Data		
Type:		Rotary
Refrigerant oil /oil charge (ml)		VG74 / 620
Input (W)		805 / 2045
Rated current (RLA/A)		4.15 / 9.30
Operation Range		
ODU	Cooling (°F / °C)	-13~122 / -25~50
	Heating (°F / °C)	-13~75 / -25~24
IDU	Cooling (°F / °C)	60~90 / 16~32
	Heating (°F / °C)	32~86 / 0~30

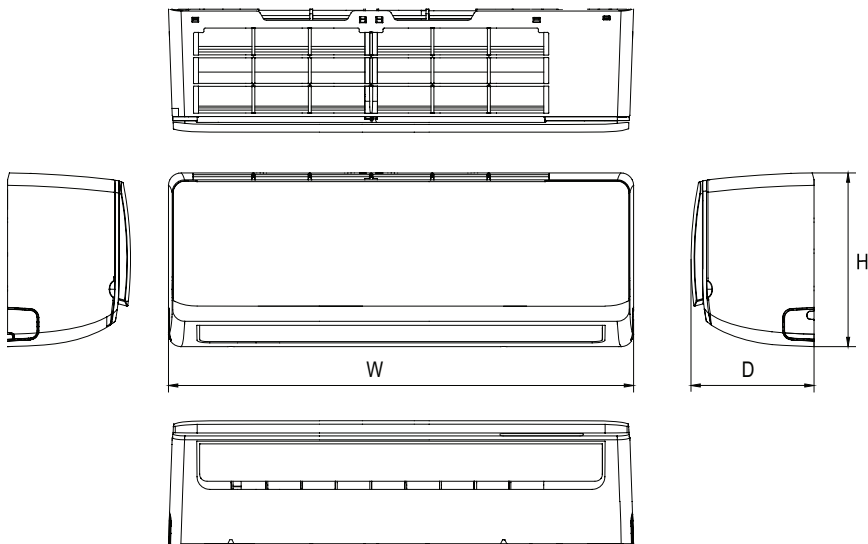
Piping

Refrigerant precharge (oz/kg)		70.55 / 2.0
Length for precharge (ft/m)		24.6 / 7.5
Liquid side (in/mm)		3/8" / Ø9.52
Gas side (in/mm)		5/8" / Ø15.9
Additional Charge of Refrigerant (oz/ft)/(g/m)		0.32 / 30
Max. refrigerant pipe length (ft/m)		164.00 / 50
Max. difference in height (ft/m)		82.00 / 25
System Data		
Refrigerant type		R454B
Design pressure (PSIG)		550 / 340
Dehumidification (pts/hr)		4.87
IDU	Fan motor model	ZKFP-58-8-1-10
	Fan motor qty (ea)	1
	Fan motor input (W)	58.0
	Fan speed (RPM)	1000 / 850 / 630
	Air flow, Turbo / H / M / L / Si (CFM)	812.3 / 618.0 / 412.0 / 317.8 / /
	Approx. Air Throw Range ft (m)	29.5 / 9
	Noise level, Turbo / H / M / L / Si (dB(A))	51.5 / 43.5 / 39.5 / 21.5 / 19
	Net weight (lbs / kg)	43.65 / 19.8
	Gross weight (lbs / kg)	55.56 / 25.2
ODU	Fan motor model	ZKFN-120-10-1L
	Fan motor qty (ea)	1
	Fan motor input (W)	170
	Fan motor speed, H / M / L (RPM)	810 / 700 / 450
	Air flow, Max (CFM)	2235.3
	Noise level, Max (dB(A))	58.0
	Net weight (lbs / kg)	122.13 / 55.4
	Gross weight (lbs / kg)	132.28 / 60.0
Accessories		
BMS-WT2-XXD (8733962707)		Wired Wall Thermostat
BMS-DRY-XXD (8733969263)		24V Interface
BMS-PK3-7X1 (8733951015)		25ft Lineset
Standard Parts		
8-733-969-088		Remote Control

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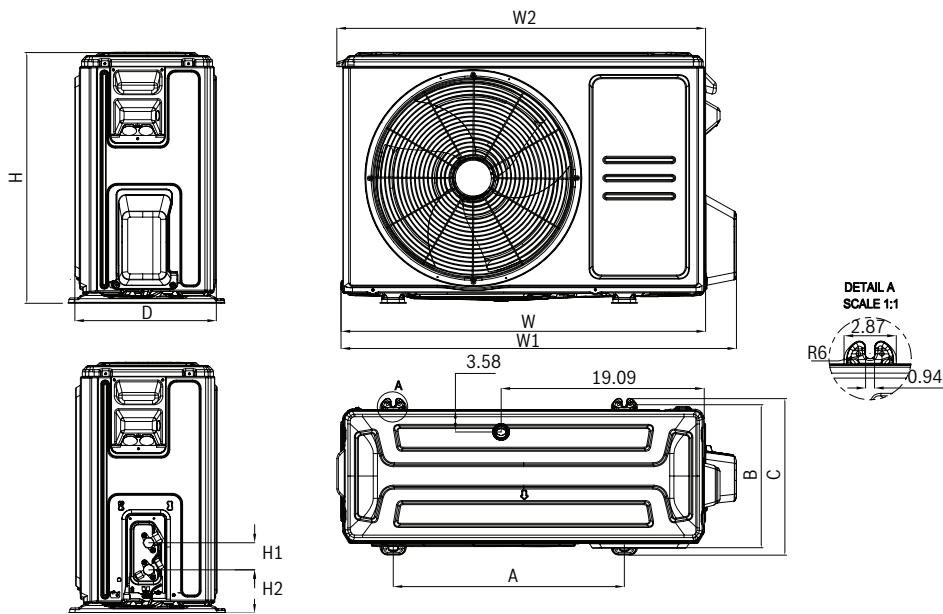


Indoor Unit Dimensions and Connections



Model	W (in)	D (in)	H (in)
BMS500-AAU024-1AHWXD	44.88	10.83	14.57

Outdoor Unit Dimensions and Connections



Model	W (in)	D (in)	H (in)	W1 (in)	W2 (in)	H1 (in)	H2 (in)	A (in)	B (in)	C (in)
BMS500-AAS024-1CSXRD	37.24	16.14	31.89	40.55	37.24	2.36	4.25	26.5	15.87	17.9

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Cooling Performance, 812 CFM																					
OD DB(°F)	ID WB(°F)	60.8°F(16.0°C)					64.4°F(18.0°C)					67°F(19.4°C)					71.6°F(22.0°C)				
	ID DB(°F)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)
-13°F (-25.0°C)	TC	23079	23068	23397	23494	24141	24079	24060	24044	24042	24022	24799	24768	24745	24741	24718	26159	26151	26098	26088	26033
	S/T	0.80	0.91	1.00	1.00	1.00	0.65	0.75	0.85	0.86	0.97	0.55	0.65	0.74	0.76	0.86	0.39	0.47	0.56	0.57	0.67
	PI	0.87	0.87	0.86	0.86	0.86	0.86	0.86	0.86	0.86	0.86	0.86	0.86	0.86	0.86	0.86	0.87	0.87	0.87	0.87	0.87
-4°F (-20.0°C)	TC	24854	24843	25197	25302	25998	25932	25910	25894	25892	25870	26707	26673	26648	26644	26619	28171	28163	28105	28095	28035
	S/T	0.80	0.92	1.00	1.00	1.00	0.65	0.76	0.85	0.87	0.98	0.55	0.65	0.74	0.76	0.86	0.39	0.48	0.56	0.57	0.67
	PI	0.90	0.90	0.90	0.89	0.89	0.89	0.89	0.89	0.89	0.89	0.89	0.89	0.89	0.89	0.89	0.90	0.90	0.90	0.90	0.90
0°F (-17.8°C)	TC	25679	25664	26140	26249	26970	26780	26751	26730	26727	26996	27567	27528	27494	27490	27477	29050	29043	29041	29045	29064
	S/T	0.81	0.93	1.00	1.00	1.00	0.66	0.77	0.86	0.88	1.00	0.55	0.66	0.75	0.77	0.87	0.39	0.48	0.57	0.58	0.68
	PI	0.90	0.90	0.90	0.90	0.90	0.90	0.90	0.90	0.90	0.90	0.90	0.90	0.90	0.90	0.91	0.91	0.91	0.91	0.91	0.92
5°F (-15.0°C)	TC	26415	26506	27112	27237	28076	27654	27661	27663	27691	28104	28538	28546	28552	28552	28590	30212	30203	30219	30222	30235
	S/T	0.82	0.94	1.00	1.00	1.00	0.66	0.78	0.87	0.88	1.00	0.56	0.67	0.76	0.78	0.88	0.39	0.48	0.57	0.59	0.69
	PI	0.91	0.91	0.91	0.91	0.92	0.92	0.92	0.92	0.92	0.92	0.92	0.92	0.92	0.92	0.92	0.94	0.94	0.94	0.94	0.94
14°F (-10.0°C)	TC	25741	25728	26253	26376	27194	27019	26995	26977	26974	27226	27945	27910	27881	27877	27849	29714	29705	29648	29638	29574
	S/T	0.81	0.93	1.00	1.00	1.00	0.65	0.76	0.86	0.87	1.00	0.55	0.66	0.75	0.76	0.87	0.39	0.48	0.56	0.57	0.67
	PI	1.05	1.05	1.04	1.04	1.04	1.04	1.04	1.04	1.04	1.04	1.04	1.04	1.04	1.04	1.04	1.05	1.05	1.05	1.05	1.05
17°F (-8.3°C)	TC	25457	25540	26130	26258	27111	26724	26735	26740	26763	27140	27636	27647	27658	27659	27693	29392	29384	29379	29385	29409
	S/T	0.82	0.93	1.00	1.00	1.00	0.66	0.77	0.87	0.88	1.00	0.55	0.66	0.76	0.77	0.87	0.39	0.48	0.57	0.58	0.68
	PI	1.08	1.08	1.08	1.08	1.08	1.08	1.08	1.08	1.08	1.08	1.08	1.08	1.08	1.08	1.08	1.09	1.09	1.09	1.09	1.09
23°F (-5.0°C)	TC	25024	25132	25782	25909	26763	26291	26299	26302	26337	26792	27198	27207	27214	27215	27259	28920	28911	28931	28935	28952
	S/T	0.83	0.95	1.00	1.00	1.00	0.66	0.78	0.88	0.89	1.00	0.56	0.67	0.76	0.78	0.89	0.39	0.48	0.57	0.59	0.69
	PI	1.05	1.05	1.06	1.06	1.06	1.06	1.06	1.06	1.06	1.06	1.07	1.07	1.07	1.07	1.07	1.09	1.09	1.09	1.09	1.09
35°F (1.7°C)	TC	24389	24390	24534	24602	25396	25529	25528	25531	25531	25599	26329	26327	26328	26329	26331	27811	27804	27803	27802	27804
	S/T	0.78	0.88	0.97	0.99	1.00	0.63	0.73	0.82	0.83	0.93	0.54	0.63	0.72	0.73	0.83	0.39	0.46	0.54	0.56	0.65
	PI	0.94	0.94	0.95	0.95	0.99	1.00	1.00	1.00	1.00	1.00	1.04	1.04	1.04	1.04	1.13	1.13	1.13	1.13	1.13	1.13
41°F (5.0°C)	TC	24579	24580	24775	24948	25685	25723	25722	25723	25723	25814	26525	26523	26525	26525	26527	28013	28006	28005	28003	28006
	S/T	0.78	0.89	0.98	1.00	1.00	0.64	0.74	0.83	0.84	0.94	0.54	0.64	0.72	0.74	0.84	0.39	0.47	0.55	0.56	0.66
	PI	0.98	0.98	0.99	1.00	1.04	1.05	1.05	1.05	1.05	1.05	1.09	1.09	1.09	1.09	1.09	1.19	1.19	1.19	1.19	1.19
47°F (8.3°C)	TC	24769	24785	25117	25230	25976	25917	25916	25917	25916	26037	26722	26719	26721	26721	26721	28218	28211	28208	28207	28210
	S/T	0.79	0.90	1.00	1.00	1.00	0.64	0.75	0.84	0.85	0.96	0.54	0.65	0.73	0.75	0.85	0.39	0.47	0.55	0.57	0.67
	PI	1.03	1.03	1.05	1.05	1.10	1.10	1.10	1.10	1.10	1.10	1.15	1.15	1.15	1.15	1.15	1.25	1.25	1.25	1.25	1.25
50°F (10.0°C)	TC	24345	25015	25859	26009	27017	25719	25733	25911	25993	27043	26722	26757	26771	26773	27030	28564	28639	28713	28722	28758
	S/T	0.90	1.00	1.00	1.00	1.00	0.71	0.85	0.96	0.99	1.00	0.59	0.72	0.83	0.85	0.99	0.40	0.50	0.60	0.62	0.74
	PI	1.18	1.18	1.18	1.18	1.18	1.18	1.18	1.18	1.18	1.18	1.18	1.18	1.18	1.18	1.18	1.17	1.17	1.17	1.17	1.17
59°F (15.0°C)	TC	24402	25258	26155	26314	27385	25819	25854	26147	26342	27412	26884	26911	26920	26960	27434	28868	28926	28983	28990	29014
	S/T	0.92	1.00	1.00	1.00	1.00	0.73	0.86	1.00	1.00	1.00	0.60	0.73	0.85	0.86	1.00	0.40	0.51	0.61	0.63	0.76
	PI	1.25	1.25	1.26	1.26	1.26	1.26	1.26	1.26	1.26	1.26	1.26	1.26	1.26	1.26	1.26	1.27	1.27	1.27	1.27	1.27
68°F (20.0°C)	TC	24448	25491	26437	26605	27734	25882	25940	26465	26633	27761	27001	27021	27070	27093	27784	29093	29147	29193	29198	29213
	S/T	0.95	1.00	1.00	1.00	1.00	0.74	0.89	1.00	1.00	1.00	0.61	0.75	0.87	0.89	1.00	0.40	0.51	0.63	0.65	0.78
	PI	1.33	1.33	1.34	1.34	1.35	1.34	1.34	1.34	1.34	1.35	1.35	1.35	1.35	1.35	1.35	1.37	1.36	1.36	1.36	1.36
77°F (25.0°C)	TC	24473	25651	26655	26832	28031	25944	26026	26684	26862	28059	27119	27149	27219	27249	28083	29290	29379	29447	29454	29481
	S/T	0.97	1.00	1.00	1.00	1.00	0.75	0.90	1.00	1.00	1.00	0.61	0.76	0.88	0.90	1.00	0.40	0.52	0.63	0.65	0.79
	PI	1.41	1.42	1.43	1.43	1.44	1.42	1.42	1.43	1.43	1.44	1.43	1.43	1.43	1.43	1.44	1.45	1.45	1.45	1.45	1.44
82°F (27.8°C)	TC	24748	26011	27060	27244	28491	26225	26326	27084	27274	28521	27445	27480	27565	27599	28546	29674	29785	29863	29871	29903
	S/T	0.99	1.00	1.00	1.00	1.00	0.76	0.91	1.00	1.00	1.00	0.62	0.77	0.89	0.92	1.00	0.40	0.52	0.64	0.66	0.80
	PI	1.59	1.59	1.60	1.60	1.61	1.60	1.60	1.60	1.60	1.61	1.61	1.60	1.60	1.60	1.61	1.62	1.62	1.62	1.62	1.62
86°F (30.0°C)	TC	25385	26652	27742	27936	29237	26771	26894	27773	27967	29267	28039	28071	28174	28215	29292	30355	30473	30545	30552	30581
	S/T	1.00	1.00	1.00	1.00	1.00	0.77	0.93	1.00	1.00	1.00	0.62	0.78	0.91	0.93	1.00	0.40	0.52	0.64	0.66	0.81
	PI	1.88	1.89	1.90	1.90	1.91	1.89	1.89	1.90	1.90	1.91	1.90	1.90	1.90	1.90	1.91	1.93	1.92	1.92	1.92	1.92
95°F (35.0°C)	TC	26634	26777	27643	27807	28906	28306	28322	28331	28365	28942	29521	29527	29545	29548	29597	31810	31798	31804	31813	31856
	S/T	0.83	0.95	1.00	1.00	1.00	0.66	0.78	0.88	0.89	1.00	0.56	0.67	0.76	0.78	0.88	0.39	0.48	0.57	0.58	0.68
	PI	2.44	2.44	2.48	2.49	2.54	2.51	2.51	2.51	2.51	2.54	2.57	2.57	2.57	2.57	2.57	2.68	2.67	2.67	2.67	2.68
104°F (40.0°C)	TC	23003	23213	24087	24232	25198	24452	2													

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Heating Performance at Indoor Dry Bulb Temperature, 812 CFM										
Outdoor DB(°F)	TCITotal Heating Capacity(BTU/hr)					PIPower Input(kW)				
	Indoor Conditions(DB °F)					Indoor Conditions(DB °F)				
	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)
-13°F(-25.0°C)	11652	11419	11362	11317	11215	2.08	2.10	2.10	2.11	2.12
-4°F(-20.0°C)	14182	13898	13829	13774	13650	2.21	2.23	2.23	2.24	2.25
0°F(-17.8°C)	15710	15396	15319	15258	15120	2.27	2.29	2.29	2.30	2.30
5°F(-15.0°C)	17620	17268	17182	17113	16959	2.34	2.36	2.36	2.37	2.38
14°F(-10.0°C)	20168	19765	19666	19588	19411	2.46	2.48	2.49	2.49	2.50
17°F(-8.3°C)	21018	20597	20494	20412	20229	2.51	2.53	2.53	2.54	2.55
22°F(-5.6°C)	22194	21750	21642	21555	21361	2.58	2.60	2.60	2.61	2.62
27°F(-2.8°C)	23371	22903	22789	22698	22493	2.65	2.67	2.68	2.68	2.69
32°F(-0.0°C)	24547	24056	23936	23840	23626	2.72	2.75	2.75	2.76	2.77
35°F(1.7°C)	25253	24748	24625	24526	24305	2.77	2.79	2.80	2.80	2.81
37°F(2.8°C)	26312	25786	25657	25554	25324	2.83	2.85	2.85	2.86	2.87
42°F(5.6°C)	28958	28379	28237	28124	27871	2.97	3.00	3.00	3.01	3.02
44.6°F(7.0°C)	30016	29416	29269	29152	28889	3.03	3.05	3.06	3.07	3.08
47°F(8.3°C)	31604	30972	30817	30694	30418	3.12	3.14	3.15	3.16	3.17
52°F(11.1°C)	31536	30905	30751	30628	30352	2.92	2.94	2.94	2.95	2.96
57°F(13.9°C)	31468	30839	30684	30562	30287	2.71	2.73	2.74	2.75	2.76
59°F(15.0°C)	31441	30812	30658	30535	30260	2.63	2.65	2.66	2.66	2.67
62°F(16.7°C)	31400	30772	30618	30496	30221	2.51	2.53	2.54	2.54	2.55
64.4°F(18.0°C)	31367	30740	30586	30464	30190	2.41	2.43	2.44	2.44	2.45
69.8°F(21.0°C)	31294	30668	30515	30392	30119	2.20	2.21	2.22	2.22	2.23
75.2°F(24.0°C)	31220	30596	30443	30321	30048	1.98	1.99	2.00	2.00	2.01